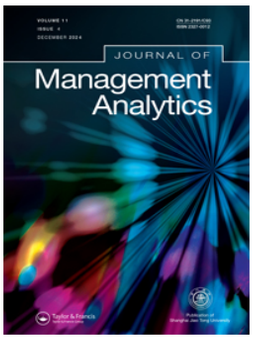

Journal of Management Analytics

ISSN: (Print) (Online) Journal homepage: www.tandfonline.com/journals/tjma20

Clustering financial institutions in countries based on a hybrid random forest and induced ordered weighted averaging

Amir Karbbasi Yazdi, Yong Tan & Paul Leger

To cite this article: Amir Karbbasi Yazdi, Yong Tan & Paul Leger (05 Mar 2025): Clustering financial institutions in countries based on a hybrid random forest and induced ordered weighted averaging, Journal of Management Analytics, DOI: 10.1080/23270012.2025.2454674

To link to this article: https://doi.org/10.1080/23270012.2025.2454674

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group

Published online: 05 Mar 2025.

Submit your article to this journal

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=tjma20

Journal of Management Analytics, 2025
https://doi.org/10.1080/23270012.2025.2454674

Clustering financial institutions in countries based on a hybrid random forest and induced ordered weighted averaging

Amir Karbbasi Yazdi[a], Yong Tan[b*] and Paul Leger[c]

[a]*Departamento de Ingeniería Industrial y de Sistemas, Facultad de Ingeniería, Universidad de Tarapacá, Arica, Chile;* [b]*School of Management, University of Bradford, Bradford, UK;* [c]*Escuela de Ingeniería, Universidad Católica del Norte, Coquimbo, Chile*

(*Received 1 July 2024; revised 11 January 2025; accepted 13 January 2025*)

The objective of this research is to assess financial institutions across various countries and regions from 2011 to 2020 using hybrid machine learning methodologies. Machine learning is employed for data analysis and prediction, proving particularly effective for extensive datasets. This paper analyzes a real bank credit dataset to examine the functionality of bank credit. The random forest method identifies Net Interest Income (gross profit and loss) as the most influential factor. Using Fuzzy C-means, we categorize the data into five clusters across the studied years. With Cluster-Induced Ordered Weighted Averaging (CIOWA), 2013 is identified as the best-performing year. This study contributes to the field by applying hybrid machine learning methods to forecast the future performance of financial institutions. Additionally, a comprehensive literature review on related issues is incorporated into this model.

Keywords: random forest; fuzzy C-means; Cluster Induced Ordered Weighted Averaging (CIOWA); credit rating performance

1. Introduction

The objective of this research is to assess financial institutions across various countries and regions from 2011 to 2020, employing hybrid machine learning methodologies (Ratnawati, 2020). Bank managers and academic researchers have focused on analyzing and predicting the causes of bank failures (Petropoulos et al., 2020). Modern institutions rely on capitalization decisions, which are an important part of the modernization process of institutions (Norgaard, 2021). There must be strict adherence by banks to national and international standards in this regard (Ferreira et al., 2019). In order to ensure the stability and solvency of the banking system, capital requirements must be met. By implementing several Basel Agreements, the capital requirements at financial institutions are periodically adjusted in accordance with economic conditions (ElBannan, 2017; Gurrea-Martínez & Remolina, 2019; Tan et al., 2024).

*Corresponding author. Email: y.tan9@bradford.ac.uk

© 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group. This is an Open Access article distributed under the terms of the Creative Commons Attribution-NonCommercial-NoDerivatives License (http://creativecommons.org/licenses/by-nc-nd/4.0/), which permits non-commercial re-use, distribution, and reproduction in any medium, provided the original work is properly cited, and is not altered, transformed, or built upon in any way. The terms on which this article has been published allow the posting of the Accepted Manuscript in a repository by the author(s) or with their consent.

Financial intermediaries are the primary source of business funding, enhancing economic growth and protecting the economy from financial crises (Haini, 2020). As financial intermediaries, banks allocate surplus funds to different borrowers. Banking plays a vital role in managing wealth and capital. This includes eradicating economic scarcity, mobilizing capital, and allocating municipal funds, among other things (Thakor, 2019). The banking system connects economic units and individuals through stock and bond markets, which are susceptible to interest rate fluctuations (Brei et al., 2020). Banking plays a significant role in financial intermediation, but it also drives the wheels of overall economic relations by generating wealth. Moreover, banks finance businesses in the real economy and ensure financial stability, both of which are enhanced by a sound banking sector. Banking stability is critical to economic growth and development. To maintain economic growth and financial stability, a country's banking system must remain robust.

Globally, many factors are considered when ranking different types of entities, including financial and non-financial factors. Country rankings, in particular, play a crucial role in shaping global dynamics and influencing governments, organizations, and individuals (Abdi et al., 2025). These rankings reflect both the strengths and weaknesses of nations. Rankings can be interpreted by examining a country's position relative to others and understanding its implications. Using fuzzy weighted average (FWA) techniques, Gavurova et al. (2022) examined bankruptcy prediction in non-financial corporations using artificial intelligence. Neural networks were used to analyze the data from 2018 and 2019. Predictive performance was assessed based on several factors, including the industry sector, scaling techniques, activation functions, and sample distribution ratios. Korol and Fotiadis (2022) used artificial neural networks, fuzzy sets, and genetic algorithms to forecast personal bankruptcies in Poland and the Taiwan region, China. Four samples were analyzed: two learning samples and two testing samples from each country and region. In each learning sample, 100 natural persons were either insolvent or solvent, while 500 households were either bankrupt or nonbankrupt.

Despite Gray Relational Analysis (GRA)'s intuitive nature, it may oversimplify the complex process of measuring sustainability. Comprehensive assessments may require multiple variables and sub-indices. Weight dispersion and unrealistic efficiency scores can result from conventional DEA models. GRA is prone to overfitting, has a 'black box' nature, and is more computationally intensive. These modelling techniques are discussed along with their advantages and disadvantages. For simpler problems, genetic algorithms may not be necessary. Evaluating fitness values over generations with genetic algorithms can be costly. Moreover, since they create linear boundaries, they don't perform well with complex or nonlinear data. Logistic Regression, while effective, can only make one type of prediction and is not suitable for continuous data analysis.

As a robust algorithm, Random Forest can handle noisy data and reduce the risk of selecting irrelevant or redundant features (Akhiat et al., 2021). Its versatility allows it to accommodate both categorical and numerical data types. Additionally, the model provides insights into the importance of features, allowing for a focus on the most informative ones. By combining predictions from several decision trees, it can improve generalization and predict performance while mitigating overfitting risks. Clustered Induced Ordered Weighted Averaging (CIOWA) is useful for data that exhibits intrinsic clustering. This tool enhances analysis that relies on grouping data points. Through clustering and ordered weighted averaging techniques, appropriate

weights are assigned to data points within clusters. This approach allows for more contextual data aggregation. Including clusters in the aggregation process can yield more accurate results, particularly when the cluster data is heterogeneous. The relationships and groupings among data points are crucial for decision-making, data fusion, and ranking tasks. Conventional clustering algorithms, such as those using the CRISP method, lead to hard partitioning of data, where each example is assigned to a single cluster (Koksalmis & Kabak, 2020).

1.1. *Motivations of study*

This paper aims to measure bank performance using a sample of banks from different countries and regions over several years. The financial performance of banks is a crucial criterion for predicting their future situations. Using the random forest algorithm, we identified the most important characteristics of the banks and ranked them in order of significance. According to the results, it can be concluded that the Net Interest Income feature (gross profit and loss) holds significant importance in predicting the amount of non-operating liabilities and losses from granted facilities. This feature plays a crucial role in determining a bank's financial performance. Non-performing loans (NPL) ranks second in importance, indicating that the amount of non-performing loans significantly impacts a bank's financial performance. By considering the importance and ranking of these features, we can assess a bank's financial performance by examining and comparing their values. For example, if a bank's Net Interest Income is high, its financial performance is likely to be strong. Conversely, if a bank has a high NPL amount, its financial performance is likely to be poor. Subsequently, banks are categorized based on the C-means model, and the similarity rate is measured.

The research questions for this study are: (1) Among the features considered, which one has the highest priority? (2) How many clusters can be categorized each year? (3) What is the similarity among clusters? (4) Which of these clusters has the best performance? By leveraging historical indicators and data, we developed a predictive model using Random Forest and CIOWA to forecast a country's future performance. Various fields, including economics, politics, and international relations, may benefit from this research. Forecasting a country's performance enables informed policy decisions, more effective resource allocation, and early identification of potential issues. Through data analytics and predictive modeling, this research aims to provide insights into a country's trajectory, leading to better-informed decision-making processes and potentially enhancing national welfare and stability. Finally, to predict corporate failure, we propose an improved hybrid Artificial Intelligence (AI) algorithm based on feature importance selection and random forest.

1.2. *Contribution*

This paper employs Random Forest (RF), Cluster-Induced Ordered Weighted Averaging (CIOWA), and fuzzy clustering approaches to model and assess the performance of financial institutions across different time periods, countries, and regions. The objective is to highlight the top-performing institutions and develop a strategic benchmarking framework. RF guides decision-makers in selecting optimal evaluation criteria. The CIOWA approach, which aggregates financial variables through

flexible weighted averaging, supports the evaluation of financial organizations by clustering financial data. The Fuzzy C-means (FCM) algorithm, a clustering method, assigns data points to multiple groups with varying degrees of membership, rather than restricting each point to a single cluster. The validity of the results has been confirmed through extensive research. The primary innovation of this work is its use of machine learning and clustering techniques to analyze financial institutions' performance across various countries, regions, and historical periods. A roadmap is developed, enabling the evaluation of financial institutions using a novel approach that integrates RF, CIOWA, and FCM. The model's credibility is further reinforced by thorough research that ensures the accuracy and reliability of the assessment results.

1.3. *Paper organization*

This paper is divided into six sections, arranged as follows. After the introduction, which outlines the research gap, contribution, and research questions, there is a literature review that discusses relevant previous studies. Section three describes the research procedure, the methods used, and the indicators employed in the current study. The methodology is presented in section four. The results of the data analysis are displayed in section five. Finally, concluding remarks are provided in the last section.

2. Literature review

Kumar et al. (2022) evaluated the performance of Indian banks. The authors developed a new model to calculate the efficiency of Indian banks and predict their level of stress using data envelopment analysis (DEA). Feature selection-based DEA is a new method that combines machine learning and feature selection to provide an alternative to traditional DEA. Assous (2022) compared selected machine learning models for predicting bank efficiency through feature selection methods with other machine learning models. In this study, they examined the main determinants of efficiency for both conventional and Islamic Saudi banks, using machine learning prediction models, including support vector machines (SVM), Chi-squared automatic interaction detectors (ChAID), linear regression, and neural networks (NN).

Sama et al. (2022) employed a multi-criteria approach to assess and rank Indian private sector banks. This paper examined the performance of Indian private sector banks based on various combinations of multi-criteria decision-making (MCDM) techniques. According to the banks' annual reports, various inputs and outputs were considered in determining each bank's performance. Several methods are used as part of the data analysis, including MCDM, Critical Incident Analysis (CRIA), Technique for Order of Preference by Similarity to Ideal Solution (TOPSIS), and grey relation analysis (GRA). Karbassi Yazdi et al. (2022) aimed to identify performance indicators and prioritize banks in Iran. A crisis, such as the COVID-19 pandemic, poses severe challenges to the banking industry. By evaluating their weaknesses, banks with flaws can be benchmarked against banks with best practices. The Delphi method was used to evaluate the data collected from a sample of Iranian banks, and it was necessary to rank the factors based on their importance. This study used the Evaluation by Area-Based Method of Ranking (EAMR), and

the Stepwise Weight Assessment Ratio Analysis (SWARA) method was employed to determine the weights for this evaluation. The paper proposed new hybrid multi-attribute decision-making (MADM) methods that are highly reliable in uncertain environments.

Wanke et al. (2022) assessed ASEAN banks' performance using MCDM and Z-Numbers. This paper examined the performance of banks in ASEAN member countries based on the CAMELS rating system, which serves as a proxy for information reliability. A comprehensive MCDM approach was developed using alternative methods to address expert preference uncertainties regarding banking ideal performance levels and relative CAMELS variable efficiency. Structured interviews were conducted to collect expert preferences, while the partial rankings of banks, based on SWARA relative efficiency weights, were calculated using fuzzy TOPSIS. Z-numbers have proven to be an effective tool for addressing the inherent fuzziness in the relationship between banking performance and financial distress, as well as the reliability of information regarding positive-ideal banking performance and CAMELS variable efficiency functions. Mirmozaffari et al. (2020) used algorithms to cluster banking systems in developing countries through DEA optimization. The primary objective of their paper was to combine machine learning with optimization to extract hidden rules, remove unrelated data from the model, and introduce the most effective Decision-Making Units (DMUs) in the optimization section, as well as the most accurate machine learning algorithm.

Petropoulos et al. (2020) used machine learning techniques to predict bank insolvencies. Their findings suggested that US-based financial institutions are more likely to go bankrupt than those in other countries. While Random Forest (RF) outperforms Neural Networks (NN) in terms of out-of-sample and out-of-time predictive performance, NN performs almost as well in out-of-time predictive performance as RF. In the broader literature, Brooks and Oikonomou (2018) assessed the performance metrics of non-financial companies in Brazil, Russia, India, China, and South Africa (BRICS) in terms of their credit risks using Logit Regression and Ordinary Least Squares in machine learning (ML) ratings.

Li et al. (2021) introduced a novel integration of clustering and optimization models, presenting a method that detects and optimizes the number of clusters to enable financial institutions to categorize data more precisely. This approach significantly enhances fraud detection and bank efficiency assessment. Petropoulos applies machine learning methods for financial management, utilizing Random Forest, SVM, Neural Networks, and conditional inference trees in conjunction with the CAMEL method (Capital Adequacy, Asset Quality, Management Quality, Earnings, and Liquidity) to evaluate these institutions. Table 1 shows the summary of previous research.

Apart from scientific articles where authors applied various artificial intelligence and machine learning methods to financial institutions, a set of literature review articles delved into the relationship between these methods and finance topics. Below, we describe some literature review articles, organized by year, on this subject. The work of Bello et al. (2023), on machine learning algorithms for credit risk assessment, explored the efficiency of clustering techniques like k-means in partitioning financial data. The paper emphasized that integrating machine learning algorithms not only enhances the classification of institutions based on credit risk but also allows for more accurate predictions of potential financial pitfalls. Doumpos and Zopounidis (2010) discussed how artificial intelligence and

Table 1. Summary of previous research.

Author(s)	Method(s)
(Brooks & Oikonomou, 2018)	Logit Regression, Ordinary Least Squares
(Petropoulos et al., 2020)	Random Forest (RF), Neural Networks (NN)
(Mirmozaffari et al., 2020)	Data Envelopment Analysis (DEA)
(Li et al., 2021)	Clustering algorithms and optimization models (e.g. k-means)
(Wanke et al., 2022)	SWARA, Fuzzy TOPSIS, Z-numbers, CAMELS
(Karbassi Yazdi et al., 2022)	Evaluation by Area-Based Method of Ranking (EAMR), Stepwise Weight Assessment Ratio Analysis (SWARA)
(Sama et al., 2022)	Critical Incident Analysis (CRIA), TOPSIS, Grey Relational Analysis (GRA)
(Assous, 2022)	Support Vector Machines (SVM), Chi-squared Automatic Interaction Detectors (ChAID), Linear Regression, Neural Networks (NN)
(Kumar & Kumar, 2022)	Data Envelopment Analysis (DEA)

operational research methods can be combined to analyze banking performance. In this paper, the authors reviewed applications of artificial intelligence methods such as Multi-Criteria Decision Analysis (MCDA) to enhance machine learning clustering techniques for evaluating financial institutions' performance. The review highlighted how these advanced methods can assess risks and efficiency more effectively than traditional models.

In Mhlanga (2021), the author presented *Financial inclusion in emerging economies: the application of machine learning and artificial intelligence in credit risk assessment*. The paper discussed how emerging economies leverage artificial intelligence and machine learning methods to assess credit risks, focusing on how clustering can identify underserved populations and provide financial institutions with tools to optimize resource allocation. Leo et al. (2019) conducted a literature review on machine learning methods in banking risk management. The authors provided an in-depth analysis of various clustering methods designed to evaluate credit risks in financial institutions. The paper illustrated how unsupervised learning techniques like hierarchical clustering and k-means are instrumental in predicting loan defaults and improving risk management strategies in the financial sector. To find the best way to allocate credits and budgets to companies, many methods were used. For example, Kumar et al. (2021) used deep learning with the k-means method to predict credit scores. Huang et al. (2024) applied k-means to detect financial fraud, and Zhu and Liu (2021) demonstrated that applying the k-means method can raise the alarm about financial risks. Another method, Factor Analysis, along with Principal Component Analysis (PCA), helps clarify issues within financial organizations. PCA screens variables to simplify groups of companies and aids in better decision-making. Rahimpour et al. (2020) applied Data Envelopment Analysis (DEA) and PCA to evaluate financial performance in organizations.

In the category of machine learning methods, additional techniques such as Visual Grouping (Zhang & Peng, 2024), Hyperplanes (Karminsky et al., 2020) and Fuzzy

Cognitive Maps (FCMs; Huang et al., 2024) have been applied. Although these methods are popular among researchers and have certain advantages, they also come with disadvantages. For instance, the k-means method requires the number of clusters to be predefined, which, in some cases, can alter the optimal solution and bias the results (Trojan et al., 2023). PCA and Factor Analysis use linear methods, but they may need nonlinear approaches to achieve better results (Gorgoglione et al., 2021). Other methods, such as Self-Organizing Maps (SOM) and Support Vector Machines (SVM), are also used. However, analyzing large amounts of data can be costly (Cervantes et al., 2020). The FCM method shows adaptability in uncertain control environments, though it does not adequately manage the degree of fuzziness (Dhanachandra & Chanu, 2020).

3. Methodology

3.1. *Random forest*

The development of the Random Forest (RF) approach was greatly influenced by the concept of Classification and Regression Trees (CART), as proposed by Breiman (2017). The Random Forest method is an ensemble machine learning approach that utilizes the CART decision tree and bootstrapping aggregation method. By employing bootstrap sampling, the Random Forest algorithm combines multiple trees by creating a tree for each bootstrap sample taken from the original dataset. As stated in the reference, there is no pruning of the individual trees. A random subset of predictor variables exclusively governs the selection of variables for each split. Every tree within the forest is constructed using an independently extracted sample, and all trees share an identical distribution. The aggregation of individual trees forms what is commonly referred to as a forest. From the complete forest, the response variable is predicted as an average (in the case of regression) or by majority vote (in the case of classification) from the predictions of all trees (Al-Hashedi & Magalingam, 2021).

Let $D = (x_1, y_1), \ldots, (x_N, y_N)$ denote the training data, where $x_i = (x_{i,1}, \ldots, x_{i,p})^T$. (1)

For $j = 1$ to J:

(1) Take a bootstrap sample D of size N.
(2) Fit a tree using binary recursive partitioning, utilizing the bootstrap sample as the training data.

    a. Initiate the process by gathering all observations into a single node.
    b. Iterate the following steps recursively for every unsplitted node until the stopping condition is satisfied.
        i. Select m predictors at random from the p available predictors.
        ii. Find the best binary split among all possible splits on the m predictors from step i.
        iii. Split the node into two descendant nodes using the chosen split from step ii.

for $k \Leftarrow 1$ to M, do
$D_k \Leftarrow \text{Bootstrap}(D)$
$h_k \Leftarrow \text{RandomTree}\left(D_k, \sqrt{f}\right)$

Forest $\Leftarrow \bigcup h_k$
for each $h_k \in$ Forest, do
for each $x_i \in D$, do
$P_{h_k}(x_i) \Leftarrow$ NodesOnPath(x_i, h_k) (2)
for each $n \in P_{h_k}(x_i)$, do
node_ $id \Leftarrow$ Get NodeID(n)
feature_id $\Leftarrow$ Hash(node_id, F)
$x'_{i,\text{ feature_}id} \Leftarrow x'_{i,\text{feature_}id} + 1$
return $D' \Leftarrow \bigcup x'_i$

D: Training data set, where each element $D = (x_1, y_1), \ldots, (x_N, y_N)$ represents the data used for training a model

$x_i = (x_{i,1}, \ldots, x_{i,p})^T$is the feature vector for the i-th observation.

J: Number of trees or iterations, where the algorithm fits multiple trees using bootstrap samples.

m: Number of predictors randomly selected from the available set of p predictors. This is used to find the best binary split among the selected predictors.

M: Number of bootstrap iterations, or the number of times a new bootstrap sample is drawn from the training data to fit a random tree.

$\boldsymbol{D_k}$: Bootstrap sample drawn from the original dataset D during iteration K.

$\boldsymbol{h_k}$: The k-th random tree created using the bootstrap sample $\left(D_k, \sqrt{f}\right)$.

$\boldsymbol{P_{h_k}(x_i)}$: Nodes on the path for observation x_i in tree h_k, representing the sequence of nodes that the observation passes through.

node_*id*: The identifier for a node in the tree.

feature_*id*: The hashed identifier of the node and the feature that is split at the nod.

3.2. *Fuzzy C-means*

The Fuzzy C-means (FCM) approach employs a fuzzy membership that assigns a degree of membership to each individual class. The relevance of the degree of membership (Dombi, 1990) in fuzzy clustering is similar to the pixel probability in a mixture modeling assumption. FCM offers the benefit of generating new clusters by considering the membership values of data points in relation to the existing classes (Ghosh & Dubey, 2013). Let us consider a set of n vectors ($X = (X_1, X_2, \ldots X_n)$ $2 < c < n$) for clustering into c groups. Each vector $x \in R^s$ is described by real-valued measurements representing the features of the object x_i. A fuzzy partition matrix, also known as a membership matrix, is employed to describe the fuzzy membership values. The set of fuzzy partition matrices (C_{xn}) is denoted by M_e and is defined as follows:

$$M_{1c} = w \in R^{cn} \mid w_{ik} \in [0, 1], \quad \forall\, 1, k; \sum_{i=1}^{c} w_{ik} = 1, \forall\, k; 0 < \sum_{k=1}^{n} w_{ik} < n, \ \forall\, i, \textit{where } 1 \le i \le c; \ 1 \le k \le n$$

The above definitions indicate that the elements have the potential to be categorized into multiple clusters, each with its own level of membership. The overall

"membership" value of an element is standardized to 1, and no single cluster can encompass all data points. The computation of the objective function in the FCM algorithm involves the utilization of membership values and Euclidean distances:

$$J_m(W, P) = \sum_{1 \le k \le n; 0 \le i \le c} (w_{ik})^m (d_{ik})^2, \tag{3}$$

where $(d_{ik})^2 = |x_k + p_i|$ is the Euclidian distance from object x to the cluster center p_i, and $m \in (1, +\infty)$ is the parameter that defines the fuzziness of the resulting clusters. The minimization of the objective function J_m through the FCM algorithm is performed by iteratively updating the partition matrix using:

$$p_i = \frac{\sum_{k=1}^{n} (w_{ik})^m x_k}{\sum_{k=1}^{n} (w_{ik})^m} \tag{4}$$

$$w_{ik}^b = \sum_{j=1}^{c} 1/[(d_{ik}^b / d_{jk}^b)^{2/m-1}] \tag{5}$$

The FCM membership function is:

$$\mu_{i,j} = \left[\sum_{t=1}^{c} \left(\frac{|x_j - v_i|_A}{|x_j - v_t|_A} \right)^{\frac{2}{m-1}} \right]^{-1}, \tag{6}$$

where $\mu_{i,j}$ is the membership value of the j-th sample in the i-th cluster. The number of clusters is represented by c, x_j is the j-th sample, and v_i is the cluster center of the i-th cluster. $||_A$ represents the norm function. The steps of the FCM algorithm are as follows:

(1) Set the number of clusters c.
(2) Choose an inner product metric, specifically the Euclidean norm, as well as the weighting metric for fuzziness.
(3) Initialize the cluster prototype $P^{(0)}$, with an iterative counter $b = 0$.
(4) Calculate the partition matrix $W^{(b)}$ using Equation (4).
(5) Update the fuzzy cluster centers $P^{(b+1)}$ using Equation (5).
(6) If $P^{(b)} - P^{(b+1)} < \varepsilon$, then stop, otherwise, repeat step 2 through 5.

X: Set of n vectors, where each vector x represents the real-valued measurements or features of an object, typically used for clustering.
M_e: Fuzzy partition matrix, representing objects' fuzzy membership values across clusters in a clustering problem.
W: Membership values in the fuzzy partition matrix, denoting how much each object belongs to each cluster.
J_m: Objective function for the Fuzzy C-Means (FCM) algorithm, which is minimized to optimize the clustering process.
d_ik: Euclidean distance between the i-th cluster center and the k-th data point.
m: Fuzziness parameter that defines how fuzzy or sharp the resulting clusters are, where $m \in (1, +\infty)$.

$\boldsymbol{p_i}$: Cluster center, calculated as a weighted mean of the data points that belong to the cluster.
$\boldsymbol{\mu_{i,j}}$: Membership value of the j-th data point in the i-th cluster.
$\boldsymbol{c}$: Number of clusters in the FCM algorithm.
$\|A\|$: Norm function used to compute distances in clustering. A represents any variable.

3.3. *OWA and Clustered Induced Ordered Weighted Averaging (CIOWA)*

Yager (1988) introduced the OWA operator, which encompasses a parameterized set of aggregation operators, including the maximum, minimum, and average variations. The operator can acquire optimal weights through the aggregation of arguments, making it extensively employed in the resolution of multi-criteria problems. An OWA operator of dimension m is a mapping, f_w: $R^m \rightarrow$R, associated with a set of weights or a weighting vector, $W = (w_1, w_2, w_3, \ldots, w_m)^T$, that meets the following requirements:

$$\sum_{i=1}^{m} w_i = 1,\ w_i \in [0, 1]$$

The function f_w is considered an m dimension OWA operator, that is:

$$f_w(a_1, a_2, a_3, \cdots, a_m) = \sum_{i=1}^{m} w_i b_i$$

where b_i represents the reordering of weight wi according to the position of a_i.

$$\begin{gathered} Min\left\{\max_{j=1,\ldots\ldots,n-1}\left|w_j - w_{j+1}\right|\right\} \\ \text{s.t.orness}\ (W) = \sum_{j=1}^{n} \frac{n-j}{n-1} \cdot w_j = \propto, 0 \leq \propto \leq 1 \\ \sum_{j=1}^{n} w_j = 1, 0 \leq w_j \leq 1, i = 1, 2, \cdots..n \end{gathered} \tag{7}$$

Yager and Filev (1999) introduced the Induced OWA (IOWA) operator, which employs an ordered inducing variable for argument rearrangement. Tuples of the specified format are aggregated using the IOWA operator. In these pairs, the value that induces the order is denoted as v_i, while the argument value is denoted as a_i. Let the $v - index$ be an index function such that $v - index(i)$ is the index of the argument pair with the i-th largest order-inducing value, then we have:

$$f_w(v_1, a_1, v_2, a_2, v_3, a_3, \cdots, v_m, a_m) = \sum_{i=1}^{m} w_i b_i = \sum_{i=1}^{m} w_i a_{v-index(i)}$$

where b_i is the value of a_i for the IOWA pair with the i-th largest v_i.

OWA Operator: Ordered Weighted Averaging operator, a mapping with associated weights that aggregates multiple inputs into a single output.

$\boldsymbol{w_i}$: Weights associated with the OWA operator, where the sum of all weights $\sum_{i=1}^{m} w_i = 1$

$\boldsymbol{b_i}$: Reordered values of the input data points based on the OWA operator.

orness (W): A measure of how much the weighting vector W favors optimistic or pessimistic aggregation in the OWA operator.

Cluster-Induced Ordered Weighted Averaging (CIOWA) has demonstrated superior performance compared to Proportional–Integral–Derivative (PID) across various dimensions in multi-criteria decision-making, especially within complex and nonlinear settings. CIOWA supports variable weighting and data classification, making it well-suited for environments characterized by subjectivity, uncertainty, and multiple goals. Unlike PID, which is primarily designed to control a single variable and respond to variations, CIOWA facilitates complex, long-term planning through anticipatory decision-making. Widely applicable across sectors and decision-making contexts, CIOWA is a highly scalable and flexible technology. Its adaptability enhances its effectiveness in managing complex interrelationships among inputs, thereby regulating their dynamics. In contrast, PID offers limited customization and is most effective for managing basic, linear systems.

3.4. *Factors*

It is important to note that the data were obtained from the World Bank. The variables used in the analysis are as follows: Financial Management (FM) encompasses a comprehensive approach to overseeing an organization's financial resources, incorporating various key indicators that shed light on different aspects of financial health and performance. Fixed Assets represent a company's long-term tangible investments, while Personnel Expenses refer to the costs associated with maintaining a workforce over time.

Financial institutions manage significant funds as part of their portfolios, with Total Deposits representing the total sum of funds they hold. Loans and Securities constitute a major portion of these funds. Financial obligations related to debt and operating costs are reflected in Total Interest Expense and Non-interest Expense, respectively. Outward FDI, GDP Growth, Political Stability, and the Absence of Violence rank are factors tied to a country's economic performance. Total Assets measure an entity's overall value, while Net Interest Income reflects the profitability of its core lending and borrowing activities. Other Interest Income is derived from sources other than interest-bearing transactions, while Non-interest Income refers to income unrelated to such transactions.

Loan Loss Provisions to Gross Loans and Non-performing Loans to Gross Loans offer insights into risk management. NPLs (Non-performing Loans) are loans at risk of default. A comprehensive view of an organization's financial landscape can be gained by analyzing the costs associated with Funds, Capital, Labor, Loans, Securities, and Non-performing Loans, which reflect the financial operations' related costs. It is important to understand that Loan Loss Provisions act as precautionary measures against potential losses. Table 2 provides the definitions of the indicators.

Table 2. Definitions of the indicators.

Indicator	Description	Measure
Outward FDI	Measures the capital a country invests in foreign markets by acquiring and establishing businesses, assets, or subsidiaries abroad	Total capital invested in foreign businesses and assets, measured in monetary terms
GDP Growth	Quantifies the rate at which a country's overall economic output, in terms of goods and services, is expanding or contracting over a specific period	Percentage change in economic output (goods and services) over a specific period, typically annually
Political Stability and Absence of Violence Rank	Assesses political stability and absence of violence in a country	Rank based on assessments of political conditions and violence levels
Financial Management (FM)	Refers to specific financial management practices within management processes, facilitating analysis and comparison across entities	Adherence to financial management practices, measured by ratios and performance metrics
Fixed Assets	Represents long-term tangible assets owned by an organization, such as land, buildings, machinery, and equipment	Total value of long-term tangible assets, as reflected in financial statements
Personnel Expenses	Encompasses costs associated with an organization's workforce, including salaries, benefits, and related expenditures	Total cost of employee compensation, including salaries and benefits
Total Deposits	Sum of all funds held by a financial institution from its customers, reflecting its ability to attract and retain deposits, crucial for liquidity and stability	Sum of all customer deposits, reflected in the institution's balance sheet
Loans	Quantifies the total amount of loans extended by a financial institution	Total value of loans extended
Securities	Refers to financial instruments such as stocks and bonds that can be bought or sold in financial markets	Value of financial instruments (e.g. stocks, bonds) held
Total Interest Expenses	Represents the overall cost incurred by an entity in paying interest on debts and financial instruments	Total cost of interest on debts and financial instruments

(*Continued*)

Table 2. Continued.

Indicator	Description	Measure
Non-interest Expenses	Includes all operating expenses incurred by an organization, excluding interest-related costs	Total operating expenses excluding interest-related costs
Total Assets	Aggregate value of all assets owned by an organization, indicating size and financial strength	Total value of all owned assets
Net Interest Income	Difference between interest earned and interest paid by a financial institution	Difference between interest earned on loans and interest paid on debts
Other Interest Income	Includes interest earnings from non-traditional sources	Interest earned from alternative sources like investments
Non-interest Income	Encompasses all revenue streams that do not result from interest-related activities	Revenue from non-interest activities, such as fees and service charges
Loan Loss Provisions to Gross Loans Ratio	Assesses the adequacy of funds set aside for potential loan losses relative to total outstanding loans	Ratio of loan loss provisions to total outstanding loans
Non-performing Loans to Gross Loans Ratio	Gauges the proportion of non-performing loans (in default or high risk) relative to total outstanding loans	Ratio of non-performing loans to total outstanding loans
NPL (Non-performing Loans)	Loans in default or at high risk of default	Total value of loans in default or at high risk
Price of Funds	Refers to the cost incurred by a financial institution to acquire funds, such as deposits and borrowings	Cost to acquire funds
Price of Capital	Reflects the cost of obtaining and maintaining capital for a business	Cost of capital
Price of Labour	Represents the total cost incurred to employ and compensate a workforce	Total cost of workforce compensation
Price of Loans	Refers to interest rates charged by a financial institution on loan products	Interest rates on loans
Price of Securities	Indicates the cost or value associated with acquiring and holding financial securities, such as stocks and bonds	Market value or acquisition cost of securities
Price of Non-performing Loans	Represents the cost or value assigned to loans in default or at high risk of default	Cost or value of non-performing loans
Loan Loss Provisions	Funds set aside by a financial institution to cover potential loan losses	Total funds reserved for potential loan losses

3.5. *Problem statement*

Evaluating banks requires a thorough performance analysis. A bank's financial stability depends on its wealth generation and management competence, which are assessed through its assets, net interest income, and financial monitoring. Risk management in banks plays a crucial role in mitigating financial risks by evaluating the likelihood of default. Labor and non-interest expenses impact operational efficiency. External factors, such as political stability and economic progress, also influence a bank's success. These components collectively contribute to improving financial ratings. The success and sustainability of institutions rely on these factors. A bank's total assets and net interest income determine its profitability and growth potential. The effectiveness of a bank's risk management strategies defines its ability to navigate defaults and recessionary periods. Personnel and non-interest expenses influence overall bank performance, while factors like political stability and GDP growth affect operations and profitability. These elements provide a comprehensive assessment of the stability and efficiency of banks.

A holistic strategy was used to evaluate a bank by aggregating multiple factors into a single score. The criteria were prioritized according to their impact on the bank's performance. Risk control and financial soundness significantly affect bank ratings. Weighting allows for ranking, and weighted scores help define performance levels. The total score enables companies to be compared based on several criteria, including financial soundness. A bank's strong performance demonstrates its operational and financial effectiveness, highlighting both strengths and weaknesses. The research by Ji et al. (2021) on the development of new induced OWA operators is particularly important, as it emphasized improving the objectivity of criteria weighting when aggregating expert opinions. We will incorporate a discussion of this development into our research, as their contributions to accurately assessing the significance of criteria offer valuable insights in the decision-making domain. Their study enhances our methodological framework, particularly in the aggregation of expert views and the weighting of criteria in multi-criteria decision-making contexts (Ji et al., 2020).

We recognize that clustering is a well-established technique, as demonstrated by Wu et al. (2005) in their study of time-course gene expression data. However, the innovation in our research stems from the specific adaptation of this methodology to our study context. Clustering was employed alongside induced OWA operators to enhance the decision-making process. In our article, we elaborate on how our clustering approach differs from traditional implementations, emphasizing its innovative integration and contribution to the overall methodology. By justifying this modification, we aim to highlight the distinctiveness of our work and its significance in advancing the field.

3.6. *Research procedure*

Phase I: Feature selection with the random forest method.
Random forests consist of 400 to 1,200 decision trees, each constructed by randomly sampling observations from the dataset and selecting features randomly.

Phase II: Clustering financial institutions with fuzzy C-means.
The technique of data clustering involves grouping a dataset into N clusters, where each data point is assigned to each cluster to a certain extent.

Phase III: Finding the best clusters and similarity rate with CIOWA.
The induced ordered weighted average (IOWA) is an aggregation operator that allows for a parameterized range of operations between the minimum and maximum.

Figure 1 shows the research procedure of this research.

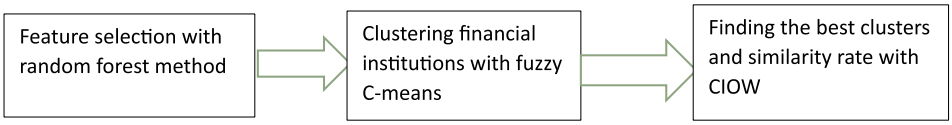

Figure 1. Research procedure.

3.7. *Validity and reliability of data*

To assess validity, we used the Singular Value Decomposition (SVD) method to evaluate the data, resulting in a Reconstruction Error of 3.6837e-07, which should ideally be close to 0. For reliability, we evaluated CIOWA by performing cross-validation, random subsampling, and data perturbation. The mean reliability score across 100 iterations was 1. Figure 2 demonstrates the reliability of CIOWA.

To assess the effectiveness of fuzzy C-means, we evaluated the Partition Coefficient (PC), which indicates the distribution of data points across clusters. This metric is defined as the summation of squared membership values of each data point to each cluster, divided by the total number of data points. Another metric, Classification Entropy (CE), measures the degree of uncertainty or

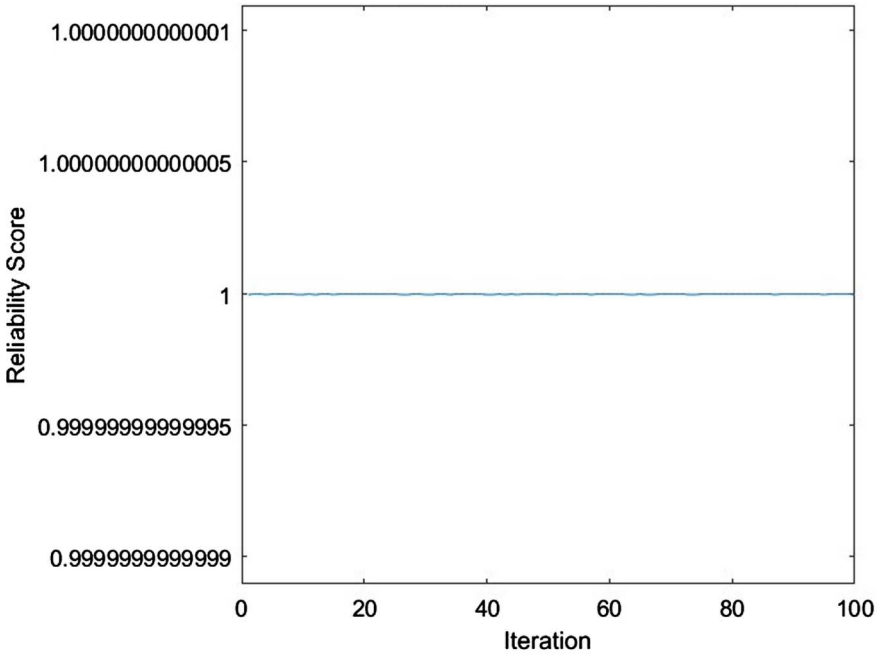

Figure 2. Reliability of CIOWA.

fuzziness present in the clustering outcomes. The Silhouette Index evaluates the degree of similarity between each data point and its assigned cluster in relation to other clusters.

A high PC value (close to 1) indicates strong and distinct clusters, while a low CE value signifies reduced uncertainty in cluster memberships – both implying dependable clustering. The Silhouette Index can also be employed to assess the level of distinctiveness among clusters, further enhancing the credibility of the results.

3.8. *Information about the data collection process*

We used a sample of banks from 43 countries and regions worldwide over the period 2011–2020. Simultaneously, data on outward foreign direct investment, GDP growth, political stability, and absence of violence were collected for these 43 countries and regions. The data on outward foreign direct investment and GDP growth were obtained from the World Bank Open Data, while political stability and absence of violence were sourced from the World Bank Worldwide Governance Indicators. The banking data was sourced from the FitchConnect database, which provides financial indicators for numerous banks across different countries and regions.

3.9. *Problem definition and research challenges*

The primary problem this study seeks to address is the accurate assessment of financial institutions' performance across various countries and regions over time, using a range of economic and financial variables. Financial institutions, particularly banks, play a crucial role in global economic stability, yet assessing their performance is complex due to the large number of variables, external factors, and the dynamic nature of financial markets. Existing literature often focuses on isolated factors affecting bank performance, such as net interest income or non-performing loans, but these fail to provide a holistic view of how various factors interact to influence performance over time.

This complexity presents several key challenges. First, the sheer volume of data across multiple countries and regions and financial indicators makes traditional analytical methods insufficient for capturing the nuanced relationships between variables. Second, the presence of noise and ambiguities in financial data requires an approach that can handle large datasets while maintaining robustness against overfitting. Third, there is a need to cluster financial institutions in a way that identifies meaningful groupings based on financial health and performance, which is difficult when traditional clustering methods treat data in isolation. Given these challenges, a more advanced machine-learning-based methodology is required to overcome the limitations of traditional financial analysis methods.

3.10. *Justification for methodology*

To address these challenges, we employed a hybrid machine learning approach that leverages Random Forest for feature selection and Cluster-Induced Ordered Weighted Averaging (CIOWA) for clustering financial institutions. Random Forest is particularly well-suited for this study as it can handle large datasets with numerous variables while mitigating the risk of overfitting through the aggregation of multiple

decision trees. This method allows us to identify the most important variables affecting bank performance, such as net interest income and non-performing loans, with greater accuracy and efficiency than traditional methods. Meanwhile, CIOWA enhances the clustering process by incorporating ordered weighted averaging, allowing us to categorize financial institutions based on their performance more effectively than traditional clustering techniques. The combination of these methods not only provides a more detailed understanding of financial performance but also ensures that the model remains robust across different years and countries, addressing the inherent complexity and variability in the data. By applying this hybrid methodology, we provide a comprehensive and accurate assessment of financial institutions' performance, offering new insights into their operational health and stability across multiple contexts.

4. Data analysis

4.1. *Order and optimization of features based on the financial performance criteria of banks*

The financial performance of banks is a crucial criterion for predicting their future situation. Therefore, using the random forest algorithm, we optimized the characteristics of the banks and ranked them in order of importance. The random forest algorithm is a machine learning method that combines several decision trees. By aggregating the results of multiple decision trees, this algorithm achieves higher accuracy and generalizability than a single decision tree. Given the characteristics of banks and their importance in determining financial performance, the random forest algorithm is the best option for ranking the importance of these characteristics due to the following features: (1) Ability to handle large datasets: the random forest algorithm can effectively process large datasets, which is crucial for banks, where financial and economic data are typically extensive and require an algorithm capable of managing such volumes. (2) Ability to deal with ambiguities and noise: The random forest algorithm can handle ambiguities and noise in the data, an essential feature in financial data analysis since financial datasets often contain noise and uncertainties that simpler algorithms may struggle to manage. (3) Ability to recognize and rank the importance of features: the random forest algorithm can automatically recognize the importance of features and rank them accordingly. This is especially significant in the context of banks, where the impact of various features on financial performance may differ. An algorithm that can assess and rank feature importance is more suitable for accurately classifying these features. Therefore, considering the characteristics of banks and the need for an algorithm that can process large datasets, handle ambiguities and noise, and rank the importance of features, the random forest algorithm is the optimal choice for this task. According to Equations 1 and 2, the feature selection process was completed. Figure 3 shows the results of the random forest method.

According to Figure 3, it can be concluded that the Net Interest Income feature (gross profit and loss) holds the highest level of importance in predicting the amount of non-performing loans (NPLGL) and the number of losses from granted facilities (LLP). This feature plays a crucial role in determining the financial performance of banks. The NPL (non-performing loans) feature is ranked second in terms of importance, indicating that the amount of non-performing loans significantly impacts a

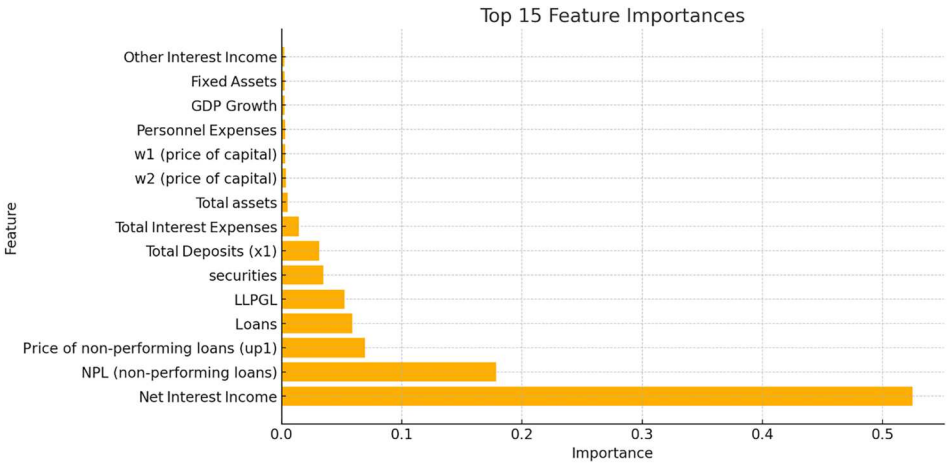

Figure 3. Random forest result.

bank's financial performance. Considering the importance and ranking of these features, it is possible to assess a bank's financial performance by examining and comparing their values. For example, if a bank's net interest income is very high, its financial performance is likely to be strong, whereas if its NPL amount is very high, its financial performance is likely to be poor.

4.2. *Clustering companies with focused attention on optimized characteristics and determining the degree of similarity among clusters*

To cluster companies according to the optimized features from the previous section, we used the fuzzy C-means algorithm. The C-means algorithm is a clustering method based on fuzzy logic, where each sample is assigned to one or more clusters, and similar samples within a cluster are positioned closer to each other. This algorithm optimizes clustering by using an objective function that minimizes the dissimilarity between samples. The C-means algorithm operates based on the following assumptions: (1) each sample is assigned to one or more clusters; (2) each sample in each cluster is assigned a fuzzy membership value; (3) the similarity between samples is calculated using Euclidean distance; (4) the goal of the algorithm is to minimize the objective function that reduces dissimilarity between samples.

The C-means algorithm works as follows: (1) determine the number of clusters and the number of samples; (2) initialize the cluster centers; (3) use the objective function to calculate the fuzzy membership of samples in the clusters; (4) update the cluster centers based on the fuzzy membership values; (5) repeat steps 3 and 4 until the cluster centers stabilize or the specified number of iterations is reached; (6) assign the samples to their respective clusters. According to Equations 3–6, fuzzy clustering has been performed. Using the C-means algorithm to cluster the financial data of banks can help in making informed financial decisions. By grouping banks into similar clusters, patterns in financial data can be identified, which can then be used in decision-making processes. After annual clustering based on the optimized features, we calculated the degree of similarity for each cluster using the IOWA method. The IOWA method calculates the degree of similarity between clusters

based on a function called the "IOWA similarity function." This function is used to calculate the similarity between two clusters using the following formula:

$$\text{similarity} = np.sum(np.minimum(cluster1, cluster2))/ \\ \text{np.sum(np.maximum(cluster1, cluster2))} \tag{8}$$

In this formula, cluster1 and cluster2 represent the two clusters whose similarity is being calculated. The functions np.minimum and np.maximum are used to compute the minimum and maximum values between two distributions, and np.sum calculates their sum. The similarity between the two clusters is then computed as the ratio of the sums of these values.

The IOWA method is considered the best for determining the degree of similarity between clusters in this study because of its two key features. First, the IOWA method fully accounts for data distribution when calculating similarity, allowing it to measure similarity comprehensively and accurately for bank financial performance assessment. By using the IOWA method, all aspects of a bank's financial performance are considered when calculating the similarity between clusters. Second, the IOWA method is sensitive to changes in the probability distributions within each cluster, paying attention to both lower and higher distributions. This sensitivity enhances the accuracy of the IOWA method in determining cluster similarity, particularly for assessing the financial performance of banks, as it can account for even small changes in the probability distributions. These two characteristics make the IOWA method the optimal choice for calculating cluster similarity in this article. Its high inference ability, greater accuracy, and higher sensitivity make it well-suited for this task. According to Equations 7 and 8, the IOWA method was implemented.

The order of importance of the features is displayed. Clustering is then conducted for each year, and the bank codes and degrees of similarity between pairs of clusters in each year are reported. Below in Figure 4, we present the similarity between pairs of clusters for 2011. The chart shows the degree of similarity, which is measured on a scale from 0 to 1, with 1 signifying strong similarity and 0 weak similarity. In 2011, clusters 0 and 4 exhibited the highest similarity in terms of financial performance and operational strategy. Clusters 0 and 3, on the other hand, had the lowest similarity, indicating significant differences in key features. Clusters 0 and 4 may have employed similar strategies or experienced comparable market conditions in 2011, while clusters 0 and 3 displayed major operational or financial differences.

Figure 5 reports the degree of similarity for each pair of clusters in 2012. This chart shows bank grouping similarities for that year. Banking was most similar in Clusters 4 and 3, suggesting a strong link between them. Clusters 1 and 4 were the least similar, indicating significant differences. The considerable similarity between clusters 4 and 3 demonstrates that banks in these clusters reacted to comparable market conditions or financial constraints in 2012, whereas the large disparity between clusters 1 and 4 reflects a variance in strategy or performance.

Figure 6 illustrates the similarity of each pair of clusters in 2013. Clusters 1 and 2, as well as Clusters 1 and 4, exhibited the most similarity in 2013, suggesting that their banks shared comparable financial or operational characteristics. In contrast, Clusters 2 and 3 showed the most significant differences, indicating considerable variation.

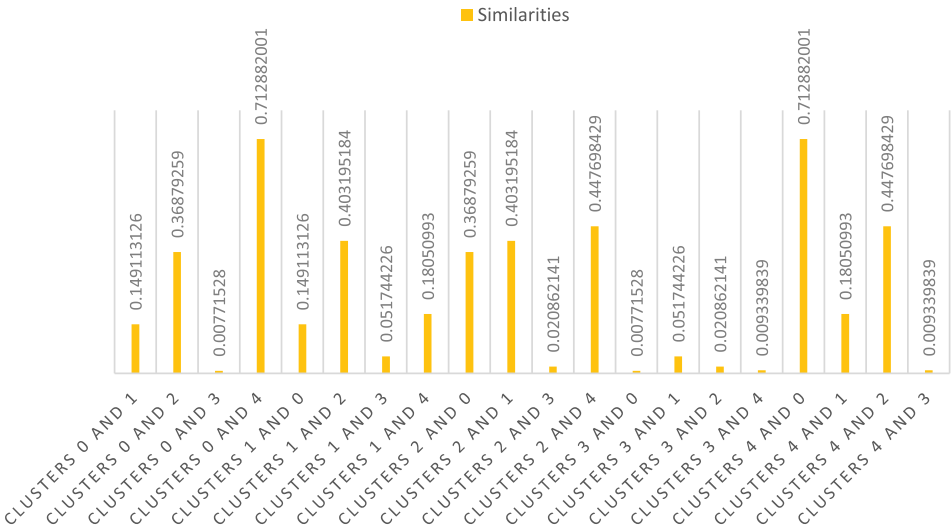

Figure 4. Similarity rate of the year 2011.

The similarities between Clusters 1 and 2 and between Clusters 1 and 4 imply that these banks employed similar techniques or encountered similar market conditions in 2013. The differences between Clusters 2 and 3 may be due to differing corporate goals or risk profiles.

Figure 7 shows the degree of similarity for each pair of clusters in 2014. In that year, Clusters 0 and 4 displayed the most similarities, indicating that their banks had comparable operations or financial indicators. The least similarity was observed between Clusters 1 and 3, suggesting significant differences. Banks in Clusters 0 and 4 may have aligned due to similar growth trajectories or operational challenges in 2014,

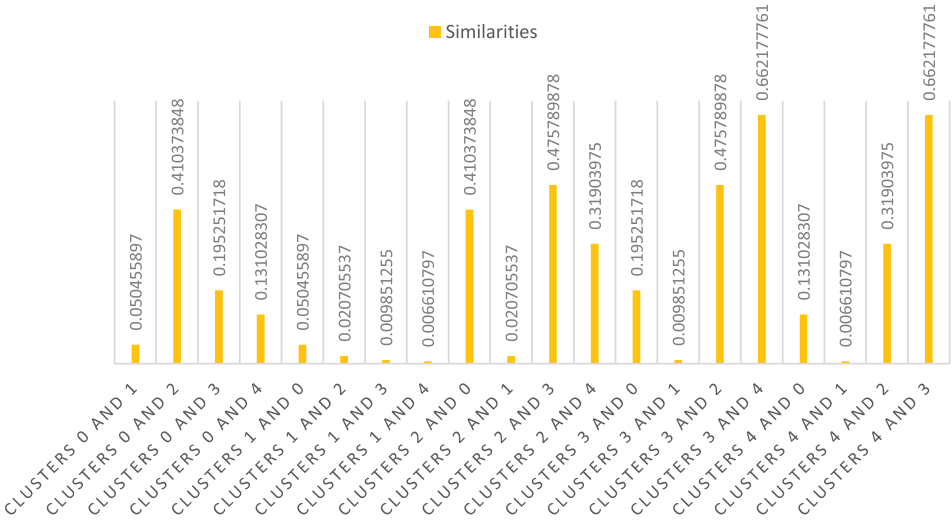

Figure 5. Similarity rate of 2012.

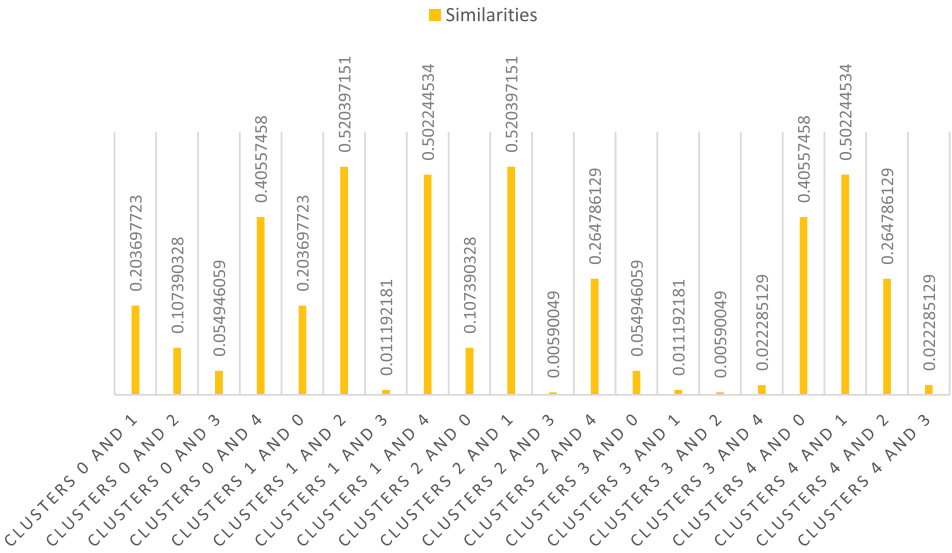

Figure 6. Similarity rate of 2013.

while banks in Clusters 1 and 3 exhibited distinct strategies, potentially driven by economic or regulatory factors.

Figure 8 reports the degree of similarity for each pair of clusters in 2015. The most comparable banks were found in Clusters 3 and 4, while banks in Clusters 1 and 2 exhibited the greatest differences. The strong similarities between Clusters 3 and 4 may imply that banks in these clusters adopted similar financial practices or risk management strategies. The lack of commonality between Clusters 1 and 2 indicates different approaches and performance during the year.

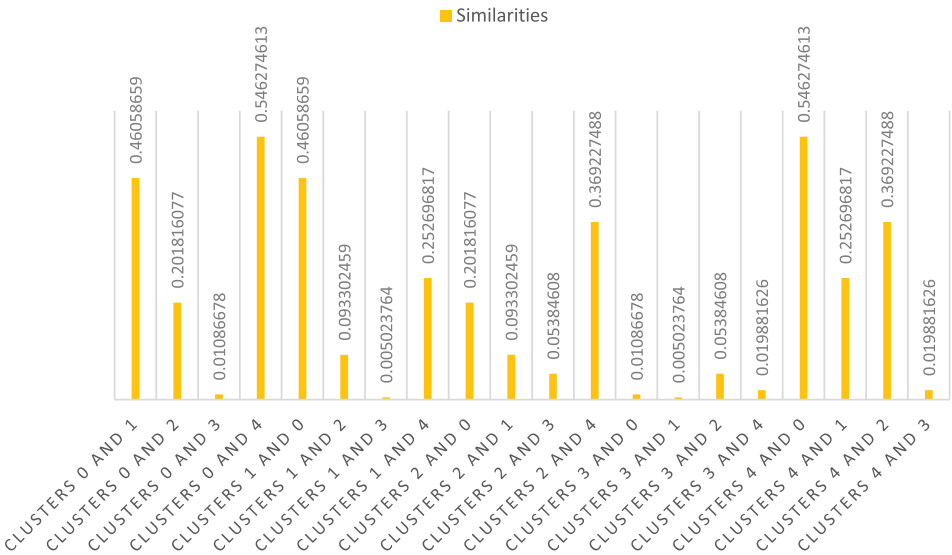

Figure 7. Similarity rate of 2014.

In 2016, Clusters 0 and 2 were the least similar, whereas Clusters 1 and 4 were the most similar. This suggests that banks in Clusters 1 and 4 shared commonalities, while Clusters 0 and 2 were more distinct. In that year, banks in Clusters 1 and 4 may have faced similar market conditions or regulatory challenges, while those in Clusters 0 and 2 might have had different market positions or corporate strategies. Figure 9 illustrates the similarity between each pair of clusters in the year 2016.

Figure 10 illustrates the similarity between each pair of clusters in 2017. Clusters 1 and 2 had the most similar banks in 2017, indicating significant agreement. Conversely, Clusters 0 and 4 showed the greatest differences. The similarity between banks in Clusters 1 and 2 likely reflects comparable financial performance due to external factors. The poor similarity between Clusters 0 and 4 suggests considerable disparities in their business models or market conditions.

In 2018, as shown in Figure 11, Clusters 0 and 1 exhibited the most similarity, indicating a strong connection between the banks. The least similar clusters, 2 and 3, demonstrated significant differences. Banks in Clusters 0 and 1 likely shared similar business strategies or challenges, whereas those in Clusters 2 and 3 adopted different market strategies.

Figure 12 reveals that Clusters 0 and 3 had the highest similarity in 2019, suggesting that their banks were quite comparable. Clusters 1 and 2 were the least similar. Banks in Clusters 0 and 3 may share operational methods or financial data, while the differences between Clusters 1 and 2 imply divergent paths for the banks.

In 2020, as depicted in Figure 13, Clusters 1 and 2 showed the highest similarity, indicating a close connection between their banks. However, Clusters 4 and 3 were the least comparable, suggesting considerable variances. The strong connection between banks in Clusters 1 and 2 may be due to similar market or regulatory responses, while the poor similarity between Clusters 4 and 3 suggests significant strategic differences.

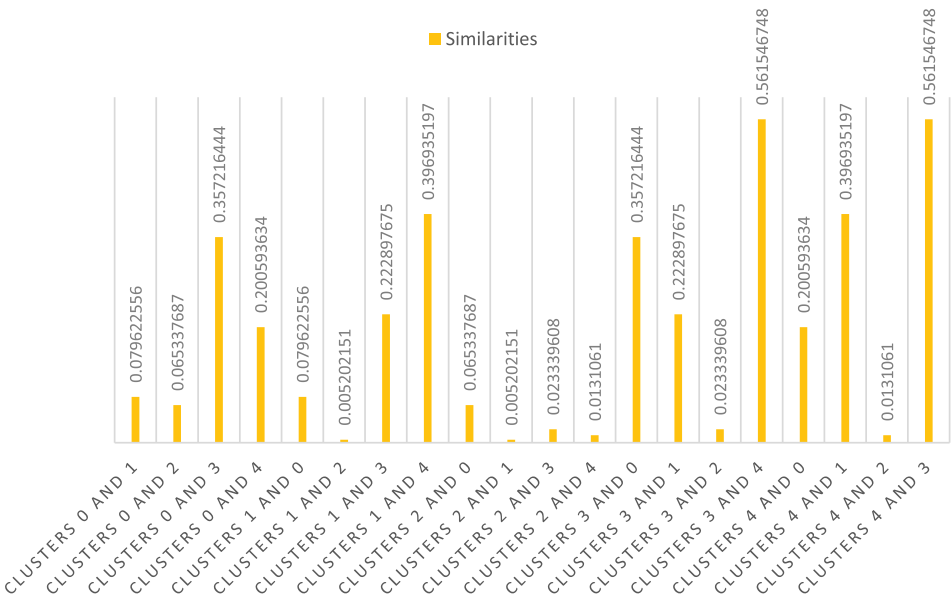

Figure 8. Similarity rate of 2015.

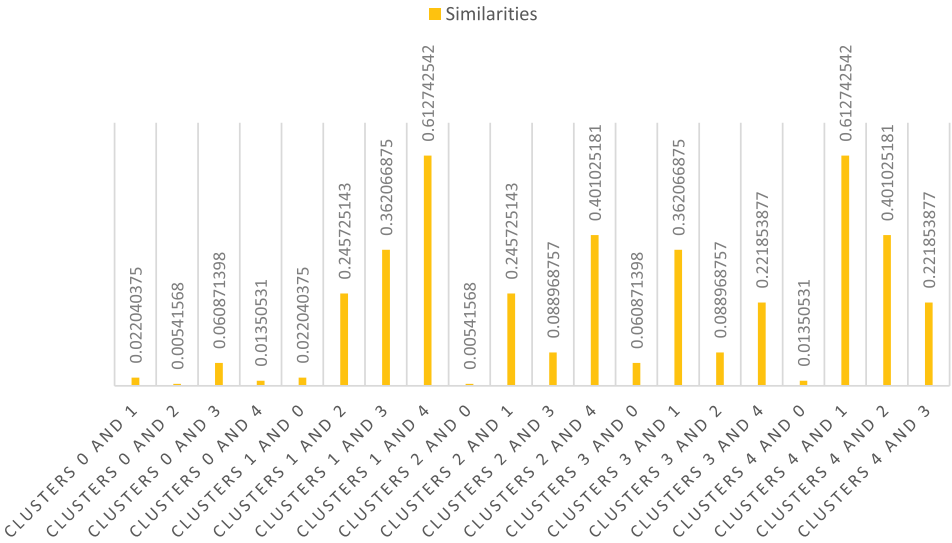

Figure 9. Similarity rate of 2016.

5. Managerial implementation

This research aims to evaluate the performance of 727 financial institutions across 44 countries from 2011 to 2020 by utilizing machine learning methods to analyze 25 key financial factors. The use of machine learning, particularly the Random Forest (RF) method, is significant because it allows us to identify and classify the most important factors influencing financial performance, even within large and complex datasets. A primary challenge lies in determining the appropriate set of criteria to assess these institutions effectively and assigning relative importance to each criterion based on its impact on performance.

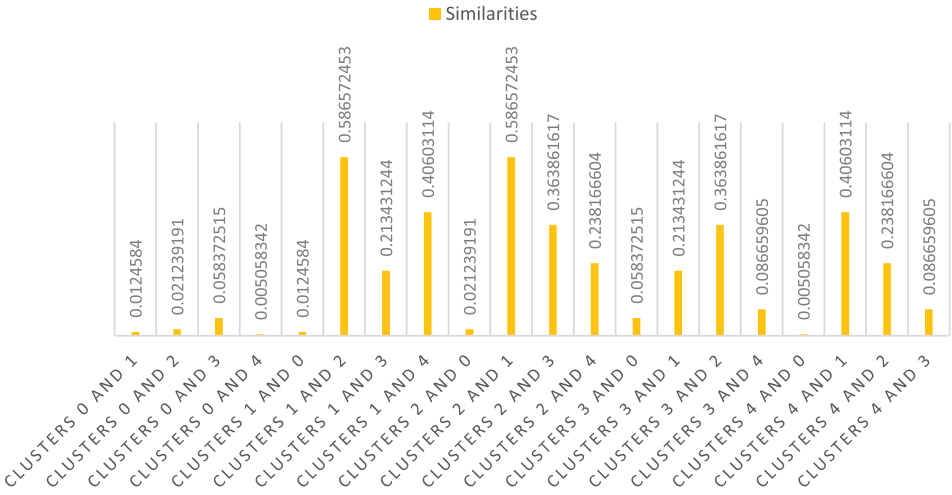

Figure 10. Similarity rate of 2017.

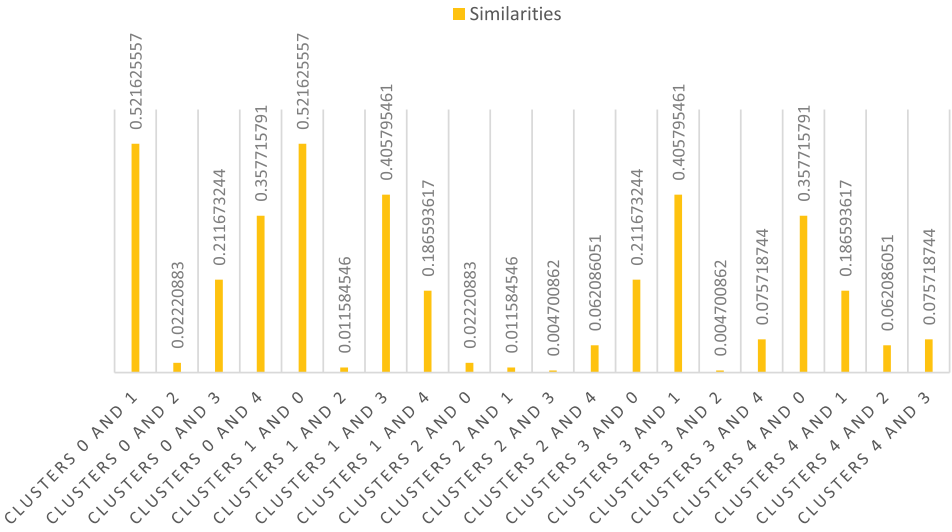

Figure 11. Similarity rate of 2018.

To address these challenges, our approach involves ranking the selected criteria according to their influence on financial stability, risk, and profitability, as identified through the RF model. By using RF, we generated an importance score for each criterion, which serves as the basis for creating an aggregate performance score for each institution. This score enables a consistent, data-driven ranking of institutions across regions and years, offering a comprehensive view of institutional performance. Our methodology thus provides a structured and transparent approach to tackling the complexities inherent in multi-criteria financial evaluations.

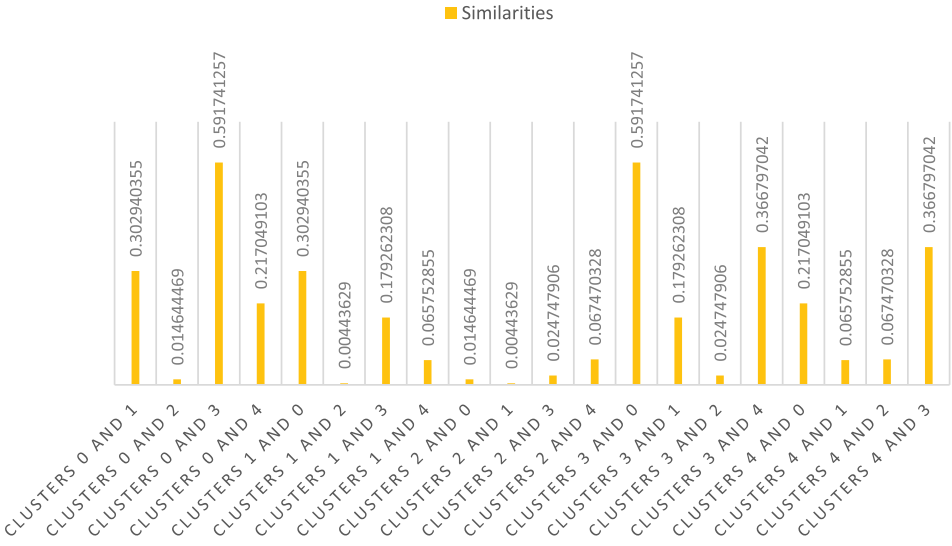

Figure 12. Similarity rate of 2019.

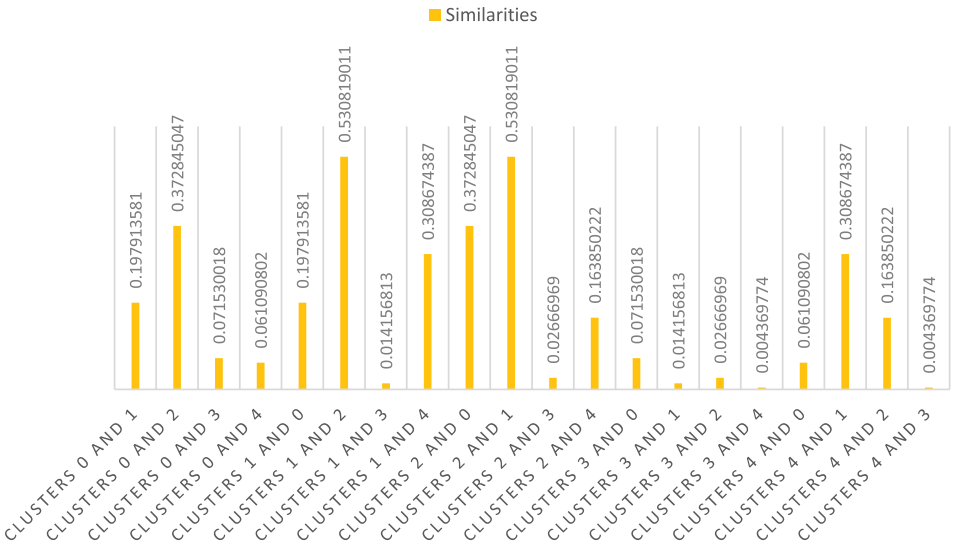

Figure 13. Similarity rate of 2020.

First, the Random Forest (RF) method was used to classify the most important factors according to their significance. The Random Forest algorithm is one of the most popular algorithms in the field of machine learning, applicable for solving classification and regression problems. Our results indicated that among the 15 factors, net interest income had the highest priority when compared with the other factors. Therefore, for evaluating the performance of these financial institutions, net interest income is of utmost importance. In contrast to negative net interest margins, positive net interest margins indicate efficient capital investment.

The institutions in a country were clustered each year using the fuzzy C-means method based on the data. It has been shown that fuzzy C-means clustering yields better results when dealing with overlapping datasets, compared to k-means clustering. Clusters are formed when the data points within a cluster are close to each other and share many common characteristics. There are various clusters of data that consist of distant points and differ from one another in terms of their characteristics. The institutions for each year were categorized into five clusters based on the number of institutions. Consequently, each cluster for each year was assessed for similarity using the IOWA method for comparison. Table 3 presents the results of the analysis from 2011 to 2020.

The table illustrates the similarities among financial institution clusters from 2011 to 2020, showing the most closely connected pairs each year. It highlights the cluster pairings with the highest similarity rates (ranging from 0 to 1, with 1 representing complete similarity) and the number of top-performing financial institutions within those clusters. Clusters 0 and 4 achieved the highest similarity score of 0.71 in 2011, while clusters 1 and 4 recorded the lowest similarity score of 0.50 in 2013. The clusters contained varying numbers of prestigious institutions, peaking at 45 in 2013 and reaching a low of 13 in both 2011 and 2012. The alignment of financial institutions' performance and operational characteristics has evolved over time due to changing market conditions and strategies, resulting in performance disparities.

Table 3. Result of data analysis.

Year	Number of Clusters	Similarity rate	Number of best Financial Institutions in each cluster
2011	0 and 4	0.71	13
2012	3 and 4	0.66	13
2013	1 and 2, 1 and 4	0.52, 0.5	45
2014	0 and 4	0.54	38
2015	3 and 4	0.56	40
2016	1 and 4	0.61	37
2017	1 and 2	0.58	36
2018	0 and 1	0.52	39
2019	0 and 3	0.59	38
2020	1 and 2	0.53	45

Based on the results of the study, it can be concluded that a greater number of financial institutions performed well in 2013, while the performance levels in 2011 and 2012 were the worst.

Clustering provides tailored guidelines for related financial organizations. Similar risk characteristics among clustered financial institutions may warrant specialized regulation, enhancing regulatory efficiency and risk management. During crises, clustered financial organizations often exhibit similar behaviors, making it beneficial for regulators to understand these clusters to prevent contagion. Grouping comparable institutions enables meaningful benchmarking by organizing financial organizations based on size, strategy, or financial health. Institutions can assess their performance by comparing practices with their peers and identifying areas for improvement. Clustering financial institutions also aids market segmentation and helps identify investment opportunities. Furthermore, clustering can highlight industry leaders and laggards in innovation and technology adoption, enhancing organizational strategies and competitiveness in dynamic financial markets. Additionally, clustering reveals country-specific traits of financial institutions, assisting in targeted problem-solving. Emerging market institutions, in particular, can benefit from clustering to address various challenges.

Recent approaches for assessing bank performance have emerged. Kumar et al. (2022) constructed a DEA model that incorporates feature selection by merging machine learning with traditional efficiency analysis. Assous (2022) examined the predictive efficacy of machine learning models – namely SVM, NN, and ChAID – in predicting bank efficiency, highlighting the importance of feature selection. Sama et al. (2022) conducted research using multi-criteria decision-making (MCDM) methodologies to assess and rank Indian banks based on various inputs and outputs. Wanke et al. (2022) evaluated ASEAN banks using MCDM and Z-numbers to address uncertainty in expert assessments. Karbassi Yazdi et al. (2022) demonstrated how hybrid Multi-Attribute Decision-Making (MADM) approaches in Iran utilize conventional mathematical techniques for financial assessment, showing enhanced precision. However, our findings differ from those of a prior study, where simpler approaches yielded similar results, possibly due to variations in sample size and data quality. Introducing a new benchmark framework enhances overall progression.

6. Conclusion

The objective of this research is to demonstrate the classification of financial institutions' performance across various countries using hybrid machine learning techniques based on economic factors. CIOWA has highlighted that net interest income is a critical determinant in evaluating financial institutions, with Random Forest (RF) playing a significant role. Furthermore, most institutions demonstrated exceptional financial performance in 2013. By addressing the existing research gap, we present a novel approach to evaluate financial institutions. This research underscores the model's capability to effectively handle intricate, multidimensional datasets, facilitating the identification of the most suitable indicators and institutions for improved performance. Implementing CIOWA and a combination of RF enables effective decision-making regarding optimal performance across countries in 2013 and the key influencing factors. Put differently, RF applies a robust approach to predict and select the most relevant features for evaluating financial indicators. Based on these factors, institutions can make informed predictions on how countries and companies within these countries can enhance their performance and use these insights as benchmarks. After a thorough analysis, decision-makers can proceed confidently with the results, guided by sensitivity analysis.

This paper does have its limitations. For instance, the model may lack structure in some areas. Additionally, this research does not account for smaller financial institutions with distinct factors. Future studies could focus on other dynamic financial factors to evaluate these institutions. Moreover, the geographical scope of this research could be expanded to include additional countries. Researchers might also explore other methods, such as Gaussian Mixture Models, for clustering algorithms.

Many institutions in today's fast-paced world face numerous challenges, one of which is that customers often fail to meet their commitments. If the number of defaults increases, these institutions might collapse due to the rising number of defaults. Consequently, institutions are eager to find solutions to address this issue. The purpose of this paper is to expand the literature on clustering financial institutions based on their performance using machine learning and CIOWA. The study uses a sample of 727 financial institutions from 44 countries worldwide. In this evaluation, 15 financial factors were analyzed, and 16 features were utilized, including Total Interest Expenses, Non-Interest Expenses, Total Assets, Net Interest Income, Other Interest Income, Non-Interest Income, Loan Loss Provisions to Gross Loans (LLPGL), Non-Performing Loans to Gross Loans (NPLGL), Non-Performing Loans (NPL), Price of Funds, Price of Capital, Price of Labor, Price of Loans, Price of Securities, Price of Non-Performing Loans, and Loan Loss Provisions (LLP).

As a machine learning method, Random Forest (RF) was employed to select features from the dataset. With this method, the importance of each feature can be determined. Subsequently, Fuzzy C-means was used to classify these financial institutions based on their performance and the importance of the features they possess, resulting in a total of five clusters. Using the CIOWA technique, the paper explains the best performance of clusters in each year based on their similarity rates obtained through CIOWA. The research demonstrates that the combination of Fuzzy C-means with RF methods is effective for optimizing cluster performance. To identify the best cluster, CIOWA was employed with several new methods. For future research, researchers should consider using other types of features, such as supervisory

methods, and explore the best clusters using other related methods like cluster analysis. Updating the dataset to the most recent year is also suggested for future studies.

In this research, we employed a subjective approach that removed human involvement to ensure consistency and objectivity in the ranking process. The use of purely objective methods for assessing the performance of financial institutions may introduce biases, potentially compromising reliability. Although this approach was suitable for the scope of our research, we recognize the potential value of incorporating expert opinions alongside objective methodologies. To enhance the depth and relevance of our findings, we will consider exploring this combination in future studies.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Data availability

The data that support the findings of this study are available from the corresponding author upon request.

References

Abdi, F., Abolmakarem, S., Yazdi, A. K., Leger, P., Tan, Y., & Coluccio, G. (2025). Predicting patients' revisit intention based on satisfaction scores: Combination of penalized regression and neural networks. *IEEE Access*, *13*, 2783–2800. https://doi.org/10.1109/ACCESS.2024.3522767

Akhiat, Y., Manzali, Y., Chahhou, M., & Zinedine, A. (2021). A New noisy random forest based method for feature selection. *Cybernetics and Information Technologies*, *21*(2), 10–28. https://doi.org/10.2478/cait-2021-0016

Al-Hashedi, K. G., & Magalingam, P. (2021). Computer Science Review.

Assous, H. F. (2022). Prediction of banks efficiency using feature selection method: Comparison between selected machine learning models. *Complexity*, *2022*, 3374489. https://doi.org/10.1155/2022/3374489

Bello, O. A., Folorunso, A., Ejiofor, O. E., Budale, F. Z., Adebayo, K., & Babatunde, O. A. (2023). Machine learning approaches for enhancing fraud prevention in financial transactions. *International Journal of Management Technology*, *10*(1), 85–108.

Brei, M., Borio, C., & Gambacorta, L. (2020). Bank intermediation activity in a low-interest-rate environment. *Economic Notes*, *49*(2), e12164. https://doi.org/10.1111/ecno.12164

Breiman, L. (2017). *Classification and regression trees*. Routledge.

Brooks, C., & Oikonomou, I. (2018). The effects of environmental, social and governance disclosures and performance on firm value: A review of the literature in accounting and finance. *The British Accounting Review*, *50*(1), 1–15. https://doi.org/10.1016/j.bar.2017.11.005

Cervantes, J., Garcia-Lamont, F., Rodríguez-Mazahua, L., & Lopez, A. (2020). A comprehensive survey on support vector machine classification: Applications, challenges and trends. *Neurocomputing*, *408*, 189–215. https://doi.org/10.1016/j.neucom.2019.10.118

Dhanachandra, N., & Chanu, Y. J. (2020). An image segmentation approach based on fuzzy c-means and dynamic particle swarm optimization algorithm. *Multimedia Tools and Applications*, *79*(25–26), 18839–18858. https://doi.org/10.1007/s11042-020-08699-8

Dombi, J. (1990). Membership function as an evaluation. *Fuzzy Sets and Systems*, *35*(1), 1–21. https://doi.org/10.1016/0165-0114(90)90014-W

Doumpos, M., & Zopounidis, C. (2010). A multicriteria decision support system for bank rating. *Decision Support Systems*, *50*(1), 55–63. https://doi.org/10.1016/j.dss.2010.07.002

ElBannan, M. A. (2017). The financial crisis, Basel accords and bank regulations: An overview. *International Journal of Accounting and Financial Reporting*, *7*(2), 225–275. https://doi.org/10.5296/ijafr.v7i2.12122

Ferreira, C., Jenkinson, N., & Wilson, C. (2019). *From Basel I to Basel III: Sequencing implementation in developing economies*. International Monetary Fund.

Gavurova, B., Jencova, S., Bacik, R., Miskufova, M., & Letkovsky, S. (2022). Artificial intelligence in predicting the bankruptcy of non-financial corporations. *Oeconomia Copernicana*, *13*(4), 1215–1251. https://doi.org/10.24136/oc.2022.035

Ghosh, S., & Dubey, S. K. (2013). Comparative analysis of k-means and fuzzy c-means algorithms. *International Journal of Advanced Computer Science and Applications*, *4*(4). https://doi.org/10.14569/IJACSA.2013.040406

Gorgoglione, A., Castro, A., Iacobellis, V., & Gioia, A. (2021). A comparison of linear and Non-linear machine learning techniques (PCA and SOM) for characterizing urban nutrient runoff. *Sustainability*, *13*(4), 2054. https://doi.org/10.3390/su13042054

Gurrea-Martínez, A., & Remolina, N. (2019). The dark side of implementing Basel capital requirements: Theory, evidence, and policy. *Journal of International Economic Law*, *22*(1), 125–152. https://doi.org/10.1093/jiel/jgz002

Haini, H. (2020). Examining the relationship between finance, institutions and economic growth: Evidence from the ASEAN economies. *Economic Change and Restructuring*, *53*(4), 519–542. https://doi.org/10.1007/s10644-019-09257-5

Huang, Z., Zheng, H., Li, C., & Che, C. (2024). Application of machine learning-based K-means clustering for financial fraud detection. *Academic Journal of Science and Technology*, *10*(1), 33–39. https://doi.org/10.54097/74414c90

Ji, C., Lu, X., & Zhang, W. (2020). A biobjective optimization model for expert opinions aggregation and its application in group decision making. *IEEE Systems Journal*, *15*(2), 2834–2844. https://doi.org/10.1109/JSYST.2020.3027716

Ji, C., Lu, X., & Zhang, W. (2021). Development of new operators for expert opinions aggregation: Average-induced ordered weighted averaging operators. *International Journal of Intelligent Systems*, *36*(2), 997–1014. https://doi.org/10.1002/int.22328

Karbassi Yazdi, A., Spulbar, C., Hanne, T., & Birau, R. (2022). Ranking performance indicators related to banking by using hybrid multicriteria methods in an uncertain environment: A case study for Iran under COVID-19 conditions. *Systems Science & Control Engineering*, *10*(1), 166–180. https://doi.org/10.1080/21642583.2022.2052996

Karminsky, A. M., Grishunin, S., Dyachkova, N., & Bisenov, M. (2020). The comparison of empirical methods for modeling credit ratings of industrial companies from BRICS countries. *Eurasian Economic Review*, *10*(2), 333–348. https://doi.org/10.1007/s40822-019-00130-4

Koksalmis, E., & Kabak, Ö. (2020). Sensor fusion based on Dempster-Shafer theory of evidence using a large scale group decision making approach. *International Journal of Intelligent Systems*, *35*(7), 1126–1162. https://doi.org/10.1002/int.22237

Korol, T., & Fotiadis, A. (2022). Implementing artificial intelligence in forecasting the risk of personal bankruptcies in Poland and Taiwan. *Oeconomia Copernicana*, *13*(2), 407–438. https://doi.org/10.24136/oc.2022.013

Kumar, A., Shanthi, D., & Bhattacharya, P. (2021). Credit score prediction system using deep learning and K-means algorithms. *Journal of Physics: Conference Series*, *1998*(1), 012027. https://doi.org/10.1088/1742-6596/1998/1/012027

Kumar, A., Shrivastav, S. K., & Mukherjee, K. (2022). Performance evaluation of Indian banks using feature selection data envelopment analysis: A machine learning perspective. *Journal of Public Affairs*, *22*(4), e2686–e2686. https://doi.org/10.1002/pa.2686

Kumar, N., & Kumar, H. (2022). A novel hybrid fuzzy time series model for prediction of COVID-19 infected cases and deaths in India. *ISA Transactions*, *124*, 69–81. https://doi.org/10.1016/j.isatra.2021.07.003

Leo, M., Sharma, S., & Maddulety, K. (2019). Machine learning in banking risk management: A literature review. *Risks*, *7*(1), 29. https://doi.org/10.3390/risks7010029

Li, B., Ding, J., Yin, Z., Li, K., Zhao, X., & Zhang, L. (2021). Optimized neural network combined model based on the induced ordered weighted averaging operator for vegetable price forecasting. *Expert Systems with Applications*, *168*, 114232. https://doi.org/10.1016/j.eswa.2020.114232

Mhlanga, D. (2021). Financial inclusion in emerging economies: The application of machine learning and artificial intelligence in credit risk assessment. *International Journal of Financial Studies*, *9*(3), 39. https://doi.org/10.3390/ijfs9030039

Mirmozaffari, M., Boskabadi, A., Azeem, G., Massah, R., Boskabadi, E., Dolatsara, H. A., & Liravian, A. (2020). Machine learning clustering algorithms based on the DEA optimization approach for banking system in developing countries. *European Journal of Engineering and Technology Research*, *5*(6), 651–658.

Norgaard, S. P. (2021). Modernization through capitalization: The hidden costs of Ethiopia's Gibe III Dam and NESTown initiative. *Human Geography*, *14*(3), 381–395. https://doi.org/10.1177/1942778621999111

Petropoulos, A., Siakoulis, V., Stavroulakis, E., & Vlachogiannakis, N. E. (2020). Predicting bank insolvencies using machine learning techniques. *International Journal of Forecasting*, *36*(3), 1092–1113. https://doi.org/10.1016/j.ijforecast.2019.11.005

Rahimpour, K., Shirouyehzad, H., Asadpour, M., & Karbasian, M. (2020). A PCA-DEA method for organizational performance evaluation based on intellectual capital and employee loyalty. *Journal of Modelling in Management*, *15*(4), 1479–1513. https://doi.org/10.1108/JM2-03-2019-0060

Ratnawati, K. (2020). The impact of financial inclusion on economic growth, poverty, income inequality, and financial stability in Asia. *The Journal of Asian Finance, Economics and Business (JAFEB)*, *7*(10), 73–85. https://doi.org/10.13106/jafeb.2020.vol7.no10.073

Sama, H. R., Kosuri, S. V. K., & Kalvakolanu, S. (2022). Evaluating and ranking the Indian private sector banks—A multi-criteria decision-making approach. *Journal of Public Affairs*, *22*(2), e2419–e2419.

Tan, Y., Karbassi Yazdi, A., Antunes, J., Wanke, P., Gunasekaran, A., Corrêa, H. L., & Coluccio, G. (2024). The quest for an ESG country rank: A performance contribution analysis/MCDM approach. *Mathematics*, *12*(12), 1865. https://doi.org/10.3390/math12121865

Thakor, A. V. (2019). *The purpose of banking: Transforming banking for stability and economic growth*. Oxford University Press.

Trojan, F., Fernandez, P. I. R., Guerreiro, M., Biuk, L., Mohamed, M. A., Siano, P., Filho, R. F. D., Marinho, M. H. N., & Siqueira, H. V. (2023). Class thresholds Pre-definition by clustering techniques for applications of ELECTRE TRI method. *Energies*, *16*(4), 1936. https://doi.org/10.3390/en16041936

Wanke, P., Azad, M. A. K., Yazdi, A. K., Birau, F. R., & Spulbar, C. M. (2022). Revisiting camels rating system and the performance of asean banks: A comprehensive mcdm/z-numbers approach. *IEEE Access*, *10*, 54098–54109. https://doi.org/10.1109/ACCESS.2022.3171339

Wu, F.-X., Kusalik, A. J., & Zhang, W.-J. (2005). Genetic weighted K-means for large-scale clustering problems. *FLAIRS*, 864–865.

Yager, R. R. (1988). On ordered weighted averaging aggregation operators in multicriteria decisionmaking. *IEEE Transactions on Systems, Man, and Cybernetics*, *18*(1), 183–190. https://doi.org/10.1109/21.87068

Yager, R. R., & Filev, D. P. (1999). Induced ordered weighted averaging operators. *IEEE Transactions on Systems, Man and Cybernetics, Part B (Cybernetics)*, *29*(2), 141–150. https://doi.org/10.1109/3477.752789

Zhang, H., & Peng, Y. (2024). Image clustering: An unsupervised approach to categorize visual data in social science research. *Sociological Methods & Research*, *53*(3), 1534–1587. https://doi.org/10.1177/00491241221082603

Zhu, Z., & Liu, N. (2021). [Retracted] early warning of financial risk based on K-means clustering algorithm. Complexity, *2021*, 5571683. https://doi.org/10.1155/2021/5571683